Helix Thready — REST /v1 Endpoint Reference

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Rev	Date	Author	Change
1	2026-07-21	swarm (API & SDKs)	Initial draft: resource groups, conventions, maturity
2	2026-07-21	swarm (API & SDKs)	Added p95 SLO; documented `x-thready-maturity` values (now real in `openapi.yaml`); expanded idempotency scope; fixed shell-quoting in the example; linked contract-tests.md
4	2026-07-22	swarm (API & SDKs)	Completeness-critic pass: addressed the `[GAP: #8 VectorDB]` pgvector-only/ Qdrant-swap-by-config seam honestly at §2.8 and in the §3 maturity table (Search row now `#1, #8, New`).
3	2026-07-22	swarm (API & SDKs)	Added the **Event Sinks** (§2.13) and **Version** (§2.14) groups (now in `openapi.yaml`, closing evt-2/ ver-2); updated the maturity count to 20; resolved rest-1 (per-op request/response examples now live in `openapi.yaml` + sdk-examples.md).

The authoritative machine contract is openapi.yaml. This document explains **behaviour** the OpenAPI cannot fully express: resource-group semantics, pagination, idempotency, filtering, rate limits, and — critically — the **maturity of the backing subsystem** behind each group (so no group is presented as GA when the gap register says its backing module is a stub/scaffold).

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4. Gaps addressed & open items

The surface topology (edge → middleware → services) is diagrammed in the area index — see diagrams/api-surface.mmd.

1. Conventions

- **Base URL** — `https://{env}.thready.hxd3v.com/v1` (prod host has no env prefix).
- **Transport** — HTTP/3 (QUIC) + HTTP/2 fallback (`vasic-digital/http3`); Brotli/gzip.
- **AuthN/AuthZ** — every route (except `security: []` ones) needs a JWT or scoped API key; role floor via `x-required-roles`; see authn-authz.md.
- **Content type** — `application/json` (UTF-8) except asset content streams.
- **Timestamps** — RFC 3339 UTC; **ids** — UUIDv4 strings unless noted.
- **Pagination** — cursor-based. List responses are `{ "data": [...], "meta": { "next_cursor": "…"|null, "total_estimate": N|null } }`. Pass `?cursor=&limit=` (limit 1–200, default 50). Exact totals are avoided at Large scale (10k+ posts/day) — only an estimate is returned.
- **Filtering** — documented per group via query params (e.g. `channel_id`, `hashtag`, `status`). `root` may pass `account_id` to cross tenants; other roles are implicitly scoped to their account.
- **Idempotency** — unsafe POSTs accept `Idempotency-Key: <uuid>` (24 h window); see error-model.md §5. In `openapi.yaml` the header is wired on every unsafe/async POST: `createApiKey`, `registerChannel`, `syncChannel`, `triggerProcessing`, `triggerReprocessing`, `redownloadAsset`, `registerSkill`, and `ingestCallback`.
- **Rate limits** — `RateLimit-*` headers on every response; 429 + `Retry-After` on breach.
- **SLO** — Aggressive posture (Q14): REST **p95 < 150 ms** for synchronous operations (excluding `202` async triggers and asset content streaming); **semantic search < 500 ms p95**. These are asserted by the performance tests in contract-tests.md §performance.
- **Maturity** — operations whose backing subsystem is not yet GA carry an `x-thready-maturity` vendor extension in `openapi.yaml` — one of `ga` | `foundation` | `build_new` | `design` — mirroring §3 below. A non-`ga` value means the contract is stable but the implementation may `503` until the referenced gap closes.
- **Errors** — the single envelope from error-model.md. Per that doc, `401`, `429` and `500` may be returned by any authenticated operation (middleware chain) and are not re-listed per operation.
- **Async** — long operations return `202 Accepted` with a job resource and emit events; clients watch the event stream (event-bus-contract.md) rather than blocking.

Example — an authenticated, paginated, filtered list (URL quoted so the shell does not split on `&` or background the command):

```
curl -s "https://thready.hxd3v.com/v1/posts?channel_id=${CID}&status=failed&limit=100" \
  -H "Authorization: Bearer ${ACCESS}" | jq '.data[].id, .meta.next_cursor'
```

2. Resource groups

2.1 Auth

`/auth/login`, `/auth/refresh`, `/auth/logout`, `/auth/me`, `/auth/mfa/totp/{enroll,verify}`, `/auth/oauth2/authorize`, `/api-keys` (list/create), `/api-keys/{keyId}` (revoke), `/.well-known/jwks.json`. Fully specified in authn-authz.md. `login`, `refresh`, and `jwks.json` are unauthenticated; everything else needs at least `user`. API-key creation requires the key's scopes to be a subset of the caller's (403 otherwise).

2.2 Accounts

`GET/POST /accounts`, `GET/PATCH/DELETE /accounts/{accountId}`, `PUT /accounts/{accountId}/branding`.

- `POST /accounts` — any authenticated user may create an account and becomes its `account_admin` (final request §6.1). `GET /accounts` returns all accounts for `root`, else the caller's memberships.
- `PATCH` — `account_admin`+; may set `retention_days` (per-account override; null = inherit global "keep indefinitely"). `DELETE` — `root` only.
- **Branding** — `PUT …/branding` sets white-label colors/logo/slogan; defaults to Thready/Helix Development (§8.3). Helix Development attribution persists in footers.

2.3 Users

`GET/POST /accounts/{accountId}/users` (list/invite), `GET/PATCH /users/{userId}`.

- Invite is `account_admin`+; a duplicate invite is `409 already_exists`. A user may belong to multiple accounts with different roles (`Membership[]`).
- `GET /users/{id}` — self, or an admin within the same tenant; cross-tenant reads are `403` (unless `root`). `PATCH` — `account_admin`+ within the tenant (e.g. change a user's role, disable).

2.4 Messengers & Channels

`GET /messengers`; `GET/POST /channels`, `GET/DELETE /channels/{channelId}`, `POST /channels/{channelId}/sync`.

- `GET /messengers` returns each platform's capabilities **and maturity** (see §3): Telegram (`foundation` — MTProto user-client reader promotion pending), Max (`build_new`).

- `POST /channels` registers a channel/group by invite link or handle (`account_admin`+); the fixtures in the final request Appendix A are valid `external_ref`s.
- `POST /channels/{id}/sync` triggers an incremental poll or a full `backfill`. **Backfill depends on the MTProto reader (Telegram) / the Max adapter (both BUILD/BUILD-NEW)** — until those land it returns `x-thready-maturity: build_new` semantics and may `503 unavailable`. Sync is async → `202`; watch `channel.synced` (sticky).

2.5 Posts & Threads

`GET /posts` (filter: `channel_id`, `hashtag`, `status`, `account_id`), `GET /posts/{id}`, `GET /posts/{id}/thread`.

- A **post** carries body, hashtags, derived `categories` (`ContentType[]`), extracted `links`, `asset_links`, and its `processing` state.
- `GET /posts/{id}/thread` returns the **complete post** = root + the full organic reply chain, **excluding Thready's own system replies** (`author.is_system == true` are never returned as processable). This is the core "assemble full thread context" requirement (§3.2.1) — tags are frequently added as a reply to a link-only root.
- Content categories are the 17 `ContentType` enum values (video, torrent, series, movie, research, documentary, concert, game, software, channel, playlist, music, book, comic, netflix, training, technology) — all built in parallel (Q31).

2.6 Processing

`POST /posts/{id}/process`, `POST /posts/{id}/reprocess`, `GET /posts/{id}/processing`, `GET /processing/jobs`, `POST /processing/callbacks/{provider}`.

- `POST /posts/{id}/process` enqueues onto the BackgroundTasks queue with an **idempotent single-claim** (Postgres row/advisory lock), so a post is processed **exactly once** under a `post.received` event storm (§3.3). Returns `202` + `ProcessingJob`; a concurrent claim returns `409 conflict`.
- `reprocess` (`account_admin`+) forces a fresh run even if already `succeeded` — the "client → REST API → System" refresh trigger (§3.2.3).
- **Precedence** — a post runs **every** matching Skill, ordered deterministically: `download > convert > analyze > research > reply` (§3.3). Exposed as `ProcessingJob.precedence`.
- `POST /processing/callbacks/{provider}` — inbound 3rd-party completion callbacks (HMAC auth, not JWT); specified in event-bus-contract.md §9.
- **Maturity** — the **Skill-dispatch engine is BUILD-NEW** atop `helix_skills` (which is a knowledge DAG, not a run engine — `[GAP: 4.1/#6]`). The endpoints are the target contract; they are backed by the new engine, not by `helix_skills` alone.

2.7 Assets & Downloads

`GET /assets`, `GET /assets/{id}`, `GET /assets/{id}/content`, `POST /assets/{id}/redownload`, `GET /posts/{id}/assets`, `GET /downloads`.

- Client links are **never raw file paths** — `GET /assets/{id}/content` resolves through the Asset Service to a **signed, Range-capable** URL (`OpenSeekable`), returning `200` / `206` (Range) or `302` to a short-lived signed URL (§7.1). Sensitive assets are gated + encrypted.
- Assets keep the **raw original** plus web renditions (`…-web` suffix; HLS/DASH ladders, §7.3/Q36); `renditions[]` lists them. Dedup + integrity via `content_hash`.
- `POST /assets/{id}/redownload` (`account_admin`+) re-fetches a `broken` physical asset via the Download Manager → `202` + `DownloadJob`.
- `GET /posts/{id}/assets` returns `AssetLink[]` with `order_index` preserving series/playlist watch order (numeric prefixes).
- **Maturity** — Asset Service is decoupled **from Catalogizer** (`P1`, mature base); the **Download Manager is BUILD-NEW** (`P0`); Boba (torrents) + MeTube (video) are FOUNDATION and integrate via the standardized callback (§2.6). `filesystem` lacks an HTTP source today (`[GAP: 6.2]`).

2.8 Search

`POST /search`.

- Semantic / keyword / **hybrid** search over **posts and generated materials** (and optionally assets). Returns ranked `{source_id, kind, score, span, snippet}`; `source_id` is the **relational PK** the vector references (vectors store reference-only metadata; §2.1.1). SLO **< 500 ms p95** (Aggressive).
- Backed by the in-house “Lumen-style” **Semantic-search service** (`embeddings` + `vectordb`/pgvector cosine) — `[GAP: New#Semantic-search service]` (`P0` BUILD-NEW).
- **Danger zone, called out** — `[GAP: #1 HelixLLM]` HelixLLM’s **default local embedder is a non-semantic `HashEmbedder` stub**; semantic search built on it silently returns garbage. The service **must** run with `HELIX_EMBEDDING_PROVIDER=llama` (real llama.cpp embeddings) and **fail loudly** (`503 unavailable`) if the hash embedder is active in a search context. `SearchResult.embedder` echoes the active provider so callers can verify.
- **Vector backend maturity** — `[GAP: #8 VectorDB]` `vectordb` is PRODUCTION on **pgvector only** (`VERIFIED`); the Qdrant / Pinecone / Milvus adapters are “constructor + config” maturity (`FLAGGED`, not integration-tested end-to-end). Thready ships on **pgvector** (co-located with the relational store, one datastore) and treats the vector backend as a **config-swappable seam**: hardening the Qdrant backend to full pgvector parity (with ANN index tuning against the `< 500 ms p95` SLO) is a `P1` tracked item, so a Large-scale deployment can switch backends without a contract change. The `/search` response shape and the `SearchResult.embedder` / `score` / `source_id` contract are identical across backends.

2.9 Skills

`GET/POST /skills`, `GET /skills/{id}`.

- Skills are **knowledge units** in the Skill-Graph DAG (`atomic → composite → umbrella`) with typed edges (`requires`/`extends`/`composes`/`recommends`/`related_to`/`alternative_to`) and a `sort_order` for dispatch precedence; `binds_content_types` wires a Skill to the `ContentType`s that trigger it.
- `POST /skills` (`account_admin`+, scope `skills:write`) registers a Skill + edges.
- **Maturity** — `helix_skills` is FOUNDATION/MVP with an inconsistent file format and an open findings backlog (`[GAP: 4.1]`); these endpoints expose the graph, while **execution** is the separate BUILD-NEW dispatch engine (§2.6).

2.10 Billing

`GET /plans`, `GET/PUT /accounts/{id}/subscription`, `GET /accounts/{id}/usage`.

- **Subscription + metered** from day one (Q11/`[OPERATOR]`). `PUT …/subscription` (`account_admin`+) changes plan; `GET …/usage` returns metered `UsageRecord[]` (`posts_processed`, `assets_bytes`, `search_queries`, `llm_tokens`, `storage_bytes`).
- Per-plan rate-limit tiers feed the edge limiter (`[OPEN: api-2]`).

2.11 Events

`GET /events` (catalog), `GET /events/{entityType}/{entityId}/sticky` (last-value snapshot). The live streams (`/v1/events/ws`, `/v1/events/stream`) and outbound/inbound callbacks are specified in event-bus-contract.md.

2.12 Ops

`GET /healthz`, `GET /readyz` (unauthenticated, unversioned in spirit — see versioning.md §2; `/metrics` is Prometheus-scraped, not in the product contract). Backed by `observability/pkg/health`.

2.13 Event sinks

`GET/POST /event-sinks`, `GET/PATCH/DELETE /event-sinks/{sinkId}` (`account_admin`+, scope `events:read` + `accounts:admin`). Registers **outbound-webhook sinks** for server-to-server subscribers that prefer HTTP push over a persistent socket. The generated HMAC `secret` is returned **once** at creation and never again (Thready signs each delivery with it via `X-Thready-Signature`). Backed by the API-plane-owned `event_sink`/`event_sink_delivery` tables and the retry/back-off contract in event-bus-contract.md §9; modelled as the `eventDelivery` channel in asyncapi.yaml. **Maturity** — `build_new` (the Event Bus service is new). Closes `[OPEN: evt-2]`.

2.14 Version

`GET /version` (`user`+) returns build + contract identity — `api_version` (semver), `contract_hash` (sha256 of the canonical OpenAPI + proto set), `proto_package`, `build_commit`, `built_at`, and any active `deprecations` with `Sunset` dates. It exists for support diagnosability (correlate a client bug to the exact deployed contract) and is the machine complement to the deprecation policy in versioning.md §6. Closes `[OPEN: ver-2]`.

3. Backing-service maturity (no-bluff)

Per the gap register — **do not present a group as GA when its backing module is a stub/scaffold**. `x-thready-maturity` (one of `ga` | `foundation` | `build_new` | `design`) is annotated on every affected operation in `openapi.yaml` — 20 operations carry it (verify with `grep -cE '^\s+x-thready-maturity:' openapi.yaml`). The negative-control test in contract-tests.md §security fails the build if any operation whose backing row below is non-GA is missing the annotation, so the two cannot drift.

Group	Backing module(s)	Status	Gap	Contract implication
Auth / Users	`digital.vasic.auth` (+ User Service)	PRODUCTION + BUILD-NEW	`#10`, `7.2`, `New`	Contract GA; needs RS256/JWKS + RBAC layer built (see authn-authz).
Channels (Telegram)	Herald MTProto reader	FOUNDATION	`#3`, `5.1`	Live read OK; **backfill pending** reader promotion.
Channels (Max)	Max adapter	**BUILD-NEW**	`#3`, `5.1`	Endpoints exist; **503 until adapter built** (Bot API + OneMe WS port).
Processing	Skill-dispatch engine	**BUILD-NEW**	`#6`, `4.1`	`helix_skills` is a knowledge DAG; execution engine is new.
Assets (serve)	Catalogizer → Asset Service	PRODUCTION (decouple P1)	`#9`, `6.1`	Serve GA; decouple + HLS/DASH transcoder pending.
Downloads	Download Manager	**BUILD-NEW**	`#4`, `6.3`	Generic multi-protocol manager is new (P0).
Downloads (video)	MeTube	FOUNDATION	`#5`, `6.5`	Poll-only today; internal bridge until webhook lands.
Search	Semantic-search svc + embeddings + vectordb	**BUILD-NEW** + trap	`#1`, `#8`, `New`	**Must use real llama.cpp embedder**; fail loudly on hash stub. pgvector GA; Qdrant swap-by-config is `P1`.
Skills (read)	`helix_skills`	FOUNDATION	`#6`, `4.1`	Graph read OK; format/backlog caveats.
Events	Event Bus service	**BUILD-NEW**	`#5`, `New`	Client-facing wrapper over eventbus/ JetStream is new.
Billing	metering/subscription	design	`#12`	Schema fixed; provider integration in deployment pack.

4. Gaps addressed & open items

- `[GAP: #3/5.1]` Herald Telegram backfill + Max adapter maturity surfaced honestly — §2.4, §3.
- `[GAP: #6/4.1]` processing endpoints backed by a BUILD-NEW Skill-dispatch engine — §2.6, §3.
- `[GAP: #1]` search embedder trap (fail loudly on HashEmbedder) — §2.8.
- `[GAP: New]` Semantic-search, Download Manager, Event Bus, User Service as BUILD-NEW — §3.
- `[GAP: #4/6.2/6.3/6.5]` download/callback contract + maturity — §2.6, §2.7.
- **TDD** — reproduce-first RED skeletons for every behaviour in this doc (pagination, idempotency single-claim, tenant isolation, maturity annotation, search fail-loud) are in contract-tests.md, mapped to the 15 mandated test types `[CONSTITUTION §11.4.27]`.

- `[RESOLVED: rest-1]` Per-endpoint request/response **examples** now live in `openapi.yaml` (login, listPosts, triggerProcessing, search incl. the fail-loud 503, ingestCallback, getStickyEvent, event-sink create, version) and end-to-end **usage** recipes per language are in sdk-examples.md. Runnable per-language quickstart repos still ship via Docs Chain once the servers exist (`[OPEN: sdkx-2]` there).
- `[OPEN: rest-2]` `/metrics` exposure format (Prometheus text vs OTLP) is set in the deployment pack; it is out of the product `/v1` contract.

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